Leonardo A. Lugarini

Aerospace Engineer | Researcher

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Curriculum Lattes

Aerospace Engineer (B.S.) and Science and Space Technologies (PG-CTE) Master's student (M.S.) with 4+ years of experience in software development and AI/ML. Proven expertise in building and deploying C++ and Python-based systems for simulation, data processing, and quantitative analysis. Deeply skilled in AI/ML model development, including Deep Learning for complex physical systems.

Education

2024 – Present **Master of Science in Science and Space Technologies (PG-CTE)**, *Aeronautics Institute of Technology (ITA)*, São José dos Campos, Brazil
Focus: Research on aerodynamic optimization of control surfaces for hypersonic vehicles.

2020 – 2024 **Bachelor of Aerospace Engineering**, *Aeronautics Institute of Technology (ITA)*, São José dos Campos, Brazil, GPA: 8.9/10.0

Experience

Feb 2025 – Present **GNC Researcher**, *Institute of Advanced Studies (IEAv)*, São José dos Campos, Brazil
- Designed and developed a high-fidelity 6DOF simulation environment in C++ for rockets and hypersonic vehicles, focusing on hypersonic aerodynamic model analysis and flight envelope validation.
- Co-designed and deployed a team development server (Ubuntu LTS) with Gitea (Git) and Samba, improving version control and shared data access for research.
- Leading development in C/C++ and Python within a collaborative Git-based workflow, managing complex codebases for simulation and analysis.

Jul 2025 – Present **Freelance Quantitative Developer**, *Confidential Client*, Remote
- Designed and developed a full-stack quantitative analysis platform, using Python for backend logic and Dash for the interactive frontend.
- Engineered the data architecture, implementing a SQLite3 database for efficient local data management, storage, and querying.
- Built a robust data ingestion pipeline to consume, process, and store financial data from the Ivolatility API.

Apr 2023 – Dez 2024 **Intern, Volatility Table**, *Legacy Capital*, São Paulo, Brazil
- Designed, developed, and maintained full-stack data processing and quantitative analysis tools using Python (Pandas, SQLAlchemy, Dash), Flask, and JavaScript (AG Grid).
- Engineered and optimized data ingestion pipelines and complex SQL queries for a high-frequency volatility database.
- Developed and evaluated quantitative trading strategies using time-series algorithms and rigorous backtesting frameworks; performed parameter optimization to enhance model performance.
- Owned and refactored a legacy Python library, implementing design patterns to improve maintainability and extensibility for the quantitative research team.
- Managed CI/CD pipelines and deployment using Git and Azure DevOps in a high-stakes, deadline-driven environment.

Apr 2024 – Sep 2024 **Intern**, *ITA Space Center (CEI)*, São José dos Campos, Brazil
- Developed a Python library for processing and ingesting binary CCSDS space packets, enabling real-time telemetry analysis for the Sports Satellite Mission.
- Built a telemetry data dashboard using the library, for rapid issue debugging and system monitoring.
- Designed the initial software architecture for a CRUD application to manage the mission's satellite database.

Sep 2022 – Sep 2023 **Scientific Initiation (PIBIC) Scholarship**, *Aeronautics Institute of Technology (ITA)*, São José dos Campos, Brazil
- Conducted computational physics research, implementing numerical models with Python (SciPy) to simulate quantum photon emissions and plasmon-polariton interactions.

Key Research Project (MS Thesis)

2024 – Present **Aerodynamic Optimization of Hypersonic Vehicle Control Surfaces**
Conducting research on the optimization of control surface geometries for hypersonic vehicles, combining aerodynamic simulation and computational optimization methods to improve performance and stability in high-speed flight regimes.

Technical Skills

AI/ML	Deep Learning, ML Model Development & Evaluation, Quantitative Modeling, Data Processing, Optimization, Pandas, NumPy, SciPy, MatLab
Programming	Python, C/C++ (Advanced); SQL (Advanced); Go, HTML, CSS, VBA (Basic)
Developer Tools	Git, Git Flow, Linux, Azure DevOps, Gitea, Docker, Samba
Engineering	Fluid Simulation (SU2, GMSH), 6DOF Modeling, CCSDS Standards

Languages

English	C2 Level (Certified)
Portuguese	Native